

ALIVE File Format Specification.pdf

Document History

Version	Date	Description
1.1	14 Dec 2010	Initial version
1.2	8 Aug 2011	Added filter flags. Corrected Example File Structure.
1.3	7 Oct 2011	Updated example data and descriptions.
1.4	21 May 2013	Removed redundant bit flags in the ECG Format block
1.5	25 Sep 2018	Added block IDs for 6 leads, removed unused 'Filtered ECG Data Block'.
1.6	5 Jun 2019	Added average beat data block, repurposed 'Location' field to 'Device Data'.

Introduction

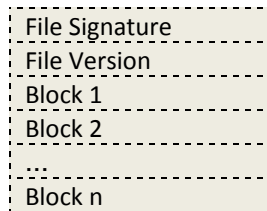
This document is intended as a guide for reading files stored by AliveCor's ECG recorder.

Overview

The ALIVE file format is a tagged file structure for storing ECG data and associated information.

File Structure

ALIVE files consist of an 8 byte signature followed by a set of "blocks". Each block consists of a block identifier, a block length, followed by the block data and a CRC checksum.



General structure of ALIVE file

File Signature

The 8 byte signature at the beginning of the file is the hex bytes: 0x41 0x4C 0x49 0x56 0x45 0x00 0x00 0x00

'A'	'L'	'I'	'V'	'E'	'0'	0	0
-----	-----	-----	-----	-----	-----	---	---

8 byte file signature

File Version

A 4 byte unsigned integer that indicates the version of the file format.

Integer	Format version
2	<=1.4
3	1.5
4	1.6

Block Structure

Blocks have the following format:

Identifier	4 bytes	An ASCII identifier for this block, e.g. "info" or "ecg".
Length	4 bytes	Length of the data block in bytes (does not include the block identifier, length or checksum)
Data	variable-sized	The block of data itself
Checksum	4 bytes	Sum of Block identifier, Block length and Block data

Checksum

The checksum is the sum of all bytes in the block, which includes the Block identifier, Block length and Block data.

Data Formats

Byte order

All multi-byte values are stored in Little-Endian (least significant byte first) order.

Strings

ALIVE files contain strings of text for specifying comments, IDs, etc. Each string field is a fixed length, with zero padding for all unused bytes. A string must be less than or equal to the available byte length for that field.

'e'	'x'	'a'	'm'	'p'	'l'	'e'	0	0	0
-----	-----	-----	-----	-----	-----	-----	---	---	---

10 Byte String Example

Info Block

The info block contains information about the ECG. Location is optional and may be zero filled.

Size	Type	Description	Example
4	UInt32	Block ID	'info'
4	UInt32	Block Data Length	264
32	String	Date Recorded. Conforms to ISO 8601 'YYYY-MM-DDThh:mm:ss.nnn±hh:mm'	2010-12-08T15:54:23.245+10:00
40	String	Recording UUID (Universally unique identifier. Automatically generated by the recording software)	58E0A7D7-EEBC-11D8-9669-0800200C9A66
44	String	Mobile Phone UDID or other phone unique identifier. Note: Apple restricted access to the phone's UDID and currently this field is left blank.	2b6f0cc904d137be2e1730235f5664094b831186
32	String	Mobile Phone Model	iPhone 4 : iPhone OS4.2
32	String	Recorder software	AliveECG v1.6.7
32	String	Recorder hardware	kardia_mobile kardia_band omron_connect triangle hwX.Y.Z fwX.Y.Zfh
52	String	Device Data (Formerly Location) Comma-separated key/value pairs	SN=AC6L100010,BAT=55
4	UInt32	Checksum	

ECG Format Block

The ECG format block contains information about the format of the ECG data. ECG data is stored unfiltered. Flags indicate the mains frequency and filters applied to stored ECG data.

Size	Type	Description	Example
4	UInt32	Block ID	'fmt '
4	UInt32	Block Data Length	8
1	BYTE	ECG Format - 0x01 = 16bit signed	1
2	UInt16	Sampling rate in Hz	300
2	UInt16	Amplitude resolution in nV	500
1	BYTE	Flags that indicate filtering of stored ECG data - Bit 0 = Polarity (Not used) - Bit 1 = Mains Frequency	8

		- o 0=50Hz - o 1=60Hz - Bit 2 = Mains Filter - Bit 3 = Low Pass Filter (Not used) - Bit 4 = 0.1Hz Baseline filter - o 0=Off - o 1=On - Bit 5 = Notch Mains Filter - Bit 6 = Enhanced Filter	
2		Reserved	
4	UInt32	Checksum	

ECG Data Block

The ecg data block contains the ECG samples for a single lead. An ATC file may contain 1 to 6 data blocks of time-aligned lead recordings. The block ID corresponds to the recording lead:

Block ID	Lead
'ecg '	Lead I
'ecg2'	Lead II
'ecg3'	Lead III
'ecg4'	aVR
'ecg5'	aVL
'ecg6'	aVF

Size	Type	Description	Example
4	UInt32	Block ID	'ecg '
4	UInt32	Block Data Length (2*n, n=number of samples of ECG data)	
2*n	Short16	n samples of ECG data	
4	UInt32	Checksum	

Average Beat Data Block

The average beat data block contains the computed average beat, generated for the specified lead. The 'Recorder Software' field in the info block identifies the version of the software used to generate the average beat.

Block ID	Lead
'avg '	Lead I Average Beat
'avg2'	Lead II Average Beat

Size	Type	Description	Example
4	UInt32	Block ID	'avg '
4	UInt32	Block Data Length (2*n, n=number of samples of ECG data)	
2*n	Short16	n samples of ECG data	
4	UInt32	Checksum	

Annotation Block

The annotation block contains a list of QRS beat detections or other annotations.

Size	Type	Description	Example
4	UInt32	Block ID	'ann '
4	UInt32	Block Data Length $4+(6*n)$, n =number of annotations.	28
4	UInt32	Tick count frequency in ticks per second.	300
6*n	UInt32	Tick count offset from start of recording	1500
	UInt16	Beat Type 0 = Unknown 1 = Normal beat 2 = Ventricular ectopic beat 3 = Atrial ectopic beat (not used)	1
4	UInt32	Checksum	

3axis Acceleration Data Block

Contains 3axis acceleration data.

Size	Type	Description	Example
4	UInt32	Block ID	'acc '
4	UInt32	Block Data Length $(20*n)$, n =number of acceleration samples.	
4	UInt32	Tick count frequency in ticks per second.	1000000
20*n	UInt64	Tick count offset, relative to the start of the ECG or start of the recording if no ECG data.	
	Float	X-axis acceleration in G's.	
	Float	Y-axis acceleration in G's.	
	Float	Z-axis acceleration in G's.	
4	UInt32	Checksum	

Example File Structure

Field	Offset	Length	Contents
ALIVE File Signature	0	8	0x41 0x4C 0x49 0x56 0x45 0x00 0x00 0x00
ALIVE File Version	8	4	2
ID	12	4	'info'
Length	16	4	264
Date Recorded	20	32	2010-12-08T15:54:23.245+10:00
Recording UUID	52	40	58E0A7D7-EEBC-11D8-9669-0800200C9A66
Mobile Phone UDID	92	44	2b6f0cc904d137be2e1730235f5664094b831186
Mobile Phone Model	136	32	iPhone 4 : iPhone OS4.2
Recorder software	168	32	AliveECG 1.6.7
Recorder hardware	200	32	19kHz 200Hz/mV
Location	232	52	geo:37.386013,-122.082932;u=35
Checksum	284	4	
ID	288	4	'fmt '
Length	282	4	8
ECG Format	296	1	1

	Sampling rate	297	2	300
	Amplitude resolution	299	2	500
	Flags	301	1	14
	Reserved	302	2	
Checksum		304	4	
ID		308	4	'ecg '
Length		312	4	18000
	ECG Sample 1	316	2	0
	ECG Sample 2	318	2	-297
	ECG Sample 3	320	2	393
	...			
	ECG Sample 9000		2	4893
Checksum			4	
ID			4	'ann '
Length			4	28
Tick frequency			4	300
	Beat 1	Tick count offset	4	1500
		Beat type	2	0
	Beat 2	Tick count offset	4	1800
		Beat type	2	1
	Beat 3	Tick count offset	4	2100
		Beat type	2	1
	Beat 4	Tick count offset	4	2400
		Beat type	2	1
Checksum			4	